Validation Report: FCA Formal Context Traceability to EO Discourse — Full Analysis

Concept: Thing | Formal context: D_Thing_Formalcontext.pdf (27 items) | Version: v3 — extended Checks 5 & 6

Check 5 extended to ALL 460 absent (— —) cells for non-placeholder items. Check 6 extended to all 96 discourse sentences.

1. Summary Table

Check	Description	Total	PASS	FAIL	Notes
Check 1	Items with explicit references	23	23	0	4 placeholders skipped
Check 2	Characteristics with explicit references	19	19	0	14 placeholders skipped
Check 3	Relationships with explicit cell references	73	73	0	
Check 4	Inferred relationships (— #)	242	242	0	
Check 5	ALL absent relationships (— —), non-placeholder items	460	460	0	All 460 cells enumerated
Check 6	ALL discourse sentence interpretations	96	96	0	96 sentences, all included

Grand total: 913 PASS, 0 FAIL across all checks.

2–4. Checks 1–3 Summary

Checks 1, 2, and 3 are identical to the v2 report. All items, characteristics, and explicit cell relationships pass. See v2 report for full detail. Results incorporated in the summary table above.

5. Check 4 Summary

All 242 inferred (— #) relationships pass. Three key reasoning corrections were applied in v2 and are retained here: (A) a plank is Composite (wood, water, organisms); (B) all Things including Abstract and Conceptual can be represented by a Conceptual thing; (C) Abstract things are in the mind (Noetic). See v2 report for full detail.

6. Check 5 — ALL Absent Relationships (— —)

All 460 absent cells across 23 non-placeholder items are enumerated below. For each, the reasoning shows why the absent mark is correct — i.e., why asserting the relationship would be unsupported by the discourse or contradicted by an EO constraint. Shared reasoning patterns arise because the same constraint (e.g., Disjoint: Concrete / Abstract) applies to all items of the same type.

Item	Absent Characteristic	Correctly Absent	Justification
56-05 [a cloud]	54-01 in a mind	YES	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
56-05 [a cloud]	54-02 can represent a Thing	YES	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
56-05 [a cloud]	- single Thing	YES	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
56-05 [a cloud]	56-29 conceived	YES	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
56-05 [a cloud]	58-03 cannot represent a Thing	YES	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
56-05 [a cloud]	56-20 Abstract thing	YES	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
56-05 [a cloud]	56-29 Intangible concrete thing	YES	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
56-05 [a cloud]	56-33 Conceptual thing	YES	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
56-05 [a cloud]	- Elemental thing	YES	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
56-05 [a cloud]	- Noetic thing	YES	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
56-05 [a cloud]	- Composite abstract thing	YES	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
56-05 [a cloud]	- Composite intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
56-05 [a cloud]	- Composite conceptual thing	YES	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
56-05 [a cloud]	- Composite noetic thing	YES	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
56-05 [a cloud]	- Elemental concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
56-05 [a cloud]	- Elemental abstract thing	YES	Concrete things are not Abstract. Not Elemental abstract.
56-05 [a cloud]	- Elemental tangible concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete:

Item	Absent Characteristic	Correctly Absent	Justification
			Composite / Elemental. Not Elemental tangible concrete.
56-05 [a cloud]	- Elemental intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
56-05 [a cloud]	- Elemental conceptual thing	**YES**	Concrete things are not Conceptual. Not Elemental conceptual.
56-05 [a cloud]	- Elemental noetic thing	**YES**	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
56-12 [a body]	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
56-12 [a body]	54-02 can represent a Thing	**YES**	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
56-12 [a body]	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
56-12 [a body]	56-29 conceived	**YES**	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
56-12 [a body]	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
56-12 [a body]	56-20 Abstract thing	**YES**	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
56-12 [a body]	56-29 Intangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
56-12 [a body]	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
56-12 [a body]	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
56-12 [a body]	- Noetic thing	**YES**	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
56-12 [a body]	- Composite abstract thing	**YES**	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
56-12 [a body]	- Composite intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
56-12 [a body]	- Composite conceptual thing	**YES**	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.

Item	Absent Characteristic	Correctly Absent	Justification
56-12 [a body]	- Composite noetic thing	YES	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
56-12 [a body]	- Elemental concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
56-12 [a body]	- Elemental abstract thing	YES	Concrete things are not Abstract. Not Elemental abstract.
56-12 [a body]	- Elemental tangible concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental tangible concrete.
56-12 [a body]	- Elemental intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
56-12 [a body]	- Elemental conceptual thing	YES	Concrete things are not Conceptual. Not Elemental conceptual.
56-12 [a body]	- Elemental noetic thing	YES	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
56-19 [a human]	54-01 in a mind	YES	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
56-19 [a human]	54-02 can represent a Thing	YES	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
56-19 [a human]	- single Thing	YES	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
56-19 [a human]	56-29 conceived	YES	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
56-19 [a human]	58-03 cannot represent a Thing	YES	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
56-19 [a human]	56-20 Abstract thing	YES	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
56-19 [a human]	56-29 Intangible concrete thing	YES	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
56-19 [a human]	56-33 Conceptual thing	YES	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
56-19 [a human]	- Elemental thing	YES	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.

Item	Absent Characteristic	Correctly Absent	Justification
56-19 [a human]	- Noetic thing	YES	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
56-19 [a human]	- Composite abstract thing	YES	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
56-19 [a human]	- Composite intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
56-19 [a human]	- Composite conceptual thing	YES	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
56-19 [a human]	- Composite noetic thing	YES	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
56-19 [a human]	- Elemental concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
56-19 [a human]	- Elemental abstract thing	YES	Concrete things are not Abstract. Not Elemental abstract.
56-19 [a human]	- Elemental tangible concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental tangible concrete.
56-19 [a human]	- Elemental intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
56-19 [a human]	- Elemental conceptual thing	YES	Concrete things are not Conceptual. Not Elemental conceptual.
56-19 [a human]	- Elemental noetic thing	YES	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
56-19 [a dog]	54-01 in a mind	YES	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
56-19 [a dog]	54-02 can represent a Thing	YES	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
56-19 [a dog]	- single Thing	YES	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
56-19 [a dog]	56-29 conceived	YES	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
56-19 [a dog]	58-03 cannot represent a Thing	YES	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
56-19 [a dog]	56-20 Abstract thing	YES	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.

Item	Absent Characteristic	Correctly Absent	Justification
56-19 [a dog]	56-29 Intangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
56-19 [a dog]	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
56-19 [a dog]	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
56-19 [a dog]	- Noetic thing	**YES**	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
56-19 [a dog]	- Composite abstract thing	**YES**	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
56-19 [a dog]	- Composite intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
56-19 [a dog]	- Composite conceptual thing	**YES**	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
56-19 [a dog]	- Composite noetic thing	**YES**	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
56-19 [a dog]	- Elemental concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
56-19 [a dog]	- Elemental abstract thing	**YES**	Concrete things are not Abstract. Not Elemental abstract.
56-19 [a dog]	- Elemental tangible concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental tangible concrete.
56-19 [a dog]	- Elemental intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
56-19 [a dog]	- Elemental conceptual thing	**YES**	Concrete things are not Conceptual. Not Elemental conceptual.
56-19 [a dog]	- Elemental noetic thing	**YES**	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
56-19 [a membership]	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
56-19 [a membership]	54-02 can represent a Thing	**YES**	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
56-19 [a membership]	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).

Item	Absent Characteristic	Correctly Absent	Justification
56-19 [a membership]	56-29 physical dimensions	**YES**	Intangible concrete things have no physical dimensions; they exist in a conceived world (56-29). Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
56-19 [a membership]	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
56-19 [a membership]	56-20 Abstract thing	**YES**	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
56-19 [a membership]	56-29 Tangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Intangible concrete things are not Tangible.
56-19 [a membership]	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
56-19 [a membership]	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
56-19 [a membership]	- Noetic thing	**YES**	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
56-19 [a membership]	- Composite abstract thing	**YES**	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
56-19 [a membership]	- Composite tangible concrete thing	**YES**	The item is Intangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite tangible concrete.
56-19 [a membership]	- Composite conceptual thing	**YES**	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
56-19 [a membership]	- Composite noetic thing	**YES**	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
56-19 [a membership]	- Elemental concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
56-19 [a membership]	- Elemental abstract thing	**YES**	Concrete things are not Abstract. Not Elemental abstract.
56-19 [a membership]	- Elemental tangible concrete thing	**YES**	The item is Intangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental tangible concrete.
56-19 [a membership]	- Elemental intangible concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental intangible concrete.
56-19 [a membership]	- Elemental conceptual thing	**YES**	Concrete things are not Conceptual. Not Elemental conceptual.
56-19 [a membership]	- Elemental noetic thing	**YES**	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
56-19 [a rental]	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.

Item	Absent Characteristic	Correctly Absent	Justification
56-19 [a rental]	54-02 can represent a Thing	YES	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
56-19 [a rental]	- single Thing	YES	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
56-19 [a rental]	56-29 physical dimensions	YES	Intangible concrete things have no physical dimensions; they exist in a conceived world (56-29). Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
56-19 [a rental]	58-03 cannot represent a Thing	YES	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
56-19 [a rental]	56-20 Abstract thing	YES	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
56-19 [a rental]	56-29 Tangible concrete thing	YES	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Intangible concrete things are not Tangible.
56-19 [a rental]	56-33 Conceptual thing	YES	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
56-19 [a rental]	- Elemental thing	YES	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
56-19 [a rental]	- Noetic thing	YES	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
56-19 [a rental]	- Composite abstract thing	YES	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
56-19 [a rental]	- Composite tangible concrete thing	YES	The item is Intangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite tangible concrete.
56-19 [a rental]	- Composite conceptual thing	YES	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
56-19 [a rental]	- Composite noetic thing	YES	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
56-19 [a rental]	- Elemental concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
56-19 [a rental]	- Elemental abstract thing	YES	Concrete things are not Abstract. Not Elemental abstract.
56-19 [a rental]	- Elemental tangible concrete thing	YES	The item is Intangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental tangible concrete.
56-19 [a rental]	- Elemental intangible concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete:

Item	Absent Characteristic	Correctly Absent	Justification
			Composite / Elemental. Not Elemental intangible concrete.
56-19 [a rental]	- Elemental conceptual thing	**YES**	Concrete things are not Conceptual. Not Elemental conceptual.
56-19 [a rental]	- Elemental noetic thing	**YES**	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
56-20 <a number>	54-02 can represent a Thing	**YES**	Abstract things do not represent other Things — they have no referent relationship (58-03). Representation is a Conceptual-thing characteristic.
56-20 <a number>	54-02 in reality	**YES**	Abstract things exist in the mind, not in reality (reasoning C: you observe 387 dogs, not 387 itself). Disjoint & complete: in a mind / in reality.
56-20 <a number>	55-12 collection of Things	**YES**	Abstract things are Elemental (not collections of component Things). A number is not decomposed into sub-Things within EO. C20 (Elemental) is marked - # for Abstract things, confirming this.
56-20 <a number>	56-29 physical dimensions	**YES**	Abstract things are purely mental constructs with no physical dimensions whatsoever.
56-20 <a number>	56-29 conceived	**YES**	"Conceived" (C10) refers to intangible Concrete things existing in a conceived world (56-29). Abstract things exist in the mind differently — they have no referent, and their being-in-the-mind is captured by C00, not C10.
56-20 <a number>	56-19 Concrete thing	**YES**	Disjoint: Concrete thing, Abstract thing. A number/boolean/date is Abstract.
56-20 <a number>	56-29 Tangible concrete thing	**YES**	Abstract things are not Concrete at all; hence not Tangible concrete.
56-20 <a number>	56-29 Intangible concrete thing	**YES**	Abstract things are not Concrete at all; hence not Intangible concrete.
56-20 <a number>	56-33 Conceptual thing	**YES**	Abstract things are not Conceptual things. These are distinct categories in EO.
56-20 <a number>	66-05 Composite thing	**YES**	Abstract things are Elemental (not collections of sub-Things). C20 (Elemental thing) is marked - # for Abstract things, and Composite/Elemental are disjoint & complete.
56-20 <a number>	- Composite concrete thing	**YES**	Abstract things are not Concrete; hence not Composite concrete.
56-20 <a number>	- Composite abstract thing	**YES**	Abstract things are Elemental (C20=-#). Disjoint & complete: Composite / Elemental. Not Composite abstract.
56-20 <a number>	- Composite tangible concrete thing	**YES**	Abstract things are not Concrete. Not Composite tangible concrete.
56-20 <a number>	- Composite intangible concrete thing	**YES**	Abstract things are not Concrete. Not Composite intangible concrete.
56-20 <a number>	- Composite conceptual thing	**YES**	Abstract things are not Conceptual. Not Composite conceptual.
56-20 <a number>	- Composite noetic thing	**YES**	Abstract things are Elemental (C20=-#). Disjoint & complete: Composite /

Item	Absent Characteristic	Correctly Absent	Justification
			Elemental. Not Composite noetic (though Noetic and Elemental).
56-20 <a number>	- Elemental concrete thing	YES	Abstract things are not Concrete. Not Elemental concrete.
56-20 <a number>	- Elemental tangible concrete thing	YES	Abstract things are not Concrete. Not Elemental tangible concrete.
56-20 <a number>	- Elemental intangible concrete thing	YES	Abstract things are not Concrete. Not Elemental intangible concrete.
56-20 <a number>	- Elemental conceptual thing	YES	Abstract things are not Conceptual. Not Elemental conceptual.
56-20 <a boolean value>	54-02 can represent a Thing	YES	Abstract things do not represent other Things — they have no referent relationship (58-03). Representation is a Conceptual-thing characteristic.
56-20 <a boolean value>	54-02 in reality	YES	Abstract things exist in the mind, not in reality (reasoning C: you observe 387 dogs, not 387 itself). Disjoint & complete: in a mind / in reality.
56-20 <a boolean value>	55-12 collection of Things	YES	Abstract things are Elemental (not collections of component Things). A number is not decomposed into sub-Things within EO. C20 (Elemental) is marked - # for Abstract things, confirming this.
56-20 <a boolean value>	56-29 physical dimensions	YES	Abstract things are purely mental constructs with no physical dimensions whatsoever.
56-20 <a boolean value>	56-29 conceived	YES	"Conceived" (C10) refers to intangible Concrete things existing in a conceived world (56-29). Abstract things exist in the mind differently — they have no referent, and their being-in-the-mind is captured by C00, not C10.
56-20 <a boolean value>	56-19 Concrete thing	YES	Disjoint: Concrete thing, Abstract thing. A number/boolean/date is Abstract.
56-20 <a boolean value>	56-29 Tangible concrete thing	YES	Abstract things are not Concrete at all; hence not Tangible concrete.
56-20 <a boolean value>	56-29 Intangible concrete thing	YES	Abstract things are not Concrete at all; hence not Intangible concrete.
56-20 <a boolean value>	56-33 Conceptual thing	YES	Abstract things are not Conceptual things. These are distinct categories in EO.
56-20 <a boolean value>	66-05 Composite thing	YES	Abstract things are Elemental (not collections of sub-Things). C20 (Elemental thing) is marked - # for Abstract things, and Composite/Elemental are disjoint & complete.
56-20 <a boolean value>	- Composite concrete thing	YES	Abstract things are not Concrete; hence not Composite concrete.
56-20 <a boolean value>	- Composite abstract thing	YES	Abstract things are Elemental (C20=-#). Disjoint & complete: Composite / Elemental. Not Composite abstract.
56-20 <a boolean value>	- Composite tangible concrete thing	YES	Abstract things are not Concrete. Not Composite tangible concrete.
56-20 <a boolean value>	- Composite intangible concrete thing	YES	Abstract things are not Concrete. Not Composite intangible concrete.

Item	Absent Characteristic	Correctly Absent	Justification
56-20 <a boolean value>	- Composite conceptual thing	**YES**	Abstract things are not Conceptual. Not Composite conceptual.
56-20 <a boolean value>	- Composite noetic thing	**YES**	Abstract things are Elemental (C20=- #). Disjoint & complete: Composite / Elemental. Not Composite noetic (though Noetic and Elemental).
56-20 <a boolean value>	- Elemental concrete thing	**YES**	Abstract things are not Concrete. Not Elemental concrete.
56-20 <a boolean value>	- Elemental tangible concrete thing	**YES**	Abstract things are not Concrete. Not Elemental tangible concrete.
56-20 <a boolean value>	- Elemental intangible concrete thing	**YES**	Abstract things are not Concrete. Not Elemental intangible concrete.
56-20 <a boolean value>	- Elemental conceptual thing	**YES**	Abstract things are not Conceptual. Not Elemental conceptual.
56-33 [the human 'Charlie Chaplin']	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
56-33 [the human 'Charlie Chaplin']	54-02 can represent a Thing	**YES**	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
56-33 [the human 'Charlie Chaplin']	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
56-33 [the human 'Charlie Chaplin']	56-29 conceived	**YES**	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
56-33 [the human 'Charlie Chaplin']	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
56-33 [the human 'Charlie Chaplin']	56-20 Abstract thing	**YES**	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
56-33 [the human 'Charlie Chaplin']	56-29 Intangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
56-33 [the human 'Charlie Chaplin']	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
56-33 [the human 'Charlie Chaplin']	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
56-33 [the human 'Charlie Chaplin']	- Noetic thing	**YES**	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
56-33 [the human 'Charlie Chaplin']	- Composite abstract thing	**YES**	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.

Item	Absent Characteristic	Correctly Absent	Justification
56-33 [the human 'Charlie Chaplin']	- Composite intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
56-33 [the human 'Charlie Chaplin']	- Composite conceptual thing	YES	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
56-33 [the human 'Charlie Chaplin']	- Composite noetic thing	YES	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
56-33 [the human 'Charlie Chaplin']	- Elemental concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
56-33 [the human 'Charlie Chaplin']	- Elemental abstract thing	YES	Concrete things are not Abstract. Not Elemental abstract.
56-33 [the human 'Charlie Chaplin']	- Elemental tangible concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental tangible concrete.
56-33 [the human 'Charlie Chaplin']	- Elemental intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
56-33 [the human 'Charlie Chaplin']	- Elemental conceptual thing	YES	Concrete things are not Conceptual. Not Elemental conceptual.
56-33 [the human 'Charlie Chaplin']	- Elemental noetic thing	YES	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
56-33 (the human 'Charlie Chaplin')	54-02 in reality	YES	Conceptual things are in the mind (59-10). Disjoint & complete: in a mind / in reality. A Conceptual thing is not in reality.
56-33 (the human 'Charlie Chaplin')	- single Thing	YES	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
56-33 (the human 'Charlie Chaplin')	56-29 physical dimensions	YES	Conceptual things are in the mind; they have no physical dimensions.
56-33 (the human 'Charlie Chaplin')	56-29 conceived	YES	"Conceived" (C10) denotes intangible Concrete things in a conceived world. Conceptual things are in the mind (C00=- #), which is a different characteristic from the intangible-concrete "conceived world."
56-33 (the human 'Charlie Chaplin')	58-03 cannot represent a Thing	YES	"Cannot represent a Thing" is specific to Abstract things (58-03). Conceptual things actively DO represent Things (56-21) — the opposite holds.
56-33 (the human 'Charlie Chaplin')	56-19 Concrete thing	YES	Disjoint: Concrete thing, Conceptual thing. A Conceptual thing is not Concrete.
56-33 (the human 'Charlie Chaplin')	56-20 Abstract thing	YES	Conceptual things are not Abstract things. These are distinct categories in EO.
56-33 (the human 'Charlie Chaplin')	56-29 Tangible concrete thing	YES	Conceptual things are not Concrete; hence not Tangible concrete.

Item	Absent Characteristic	Correctly Absent	Justification
56-33 (the human 'Charlie Chaplin')	56-29 Intangible concrete thing	**YES**	Conceptual things are not Concrete; hence not Intangible concrete.
56-33 (the human 'Charlie Chaplin')	- Elemental thing	**YES**	The Conceptual thing is marked as Composite (C19=- #). Disjoint & complete: Composite / Elemental.
56-33 (the human 'Charlie Chaplin')	- Composite concrete thing	**YES**	Conceptual things are not Concrete. Disjoint: Concrete / Conceptual. Not Composite concrete.
56-33 (the human 'Charlie Chaplin')	- Composite abstract thing	**YES**	Conceptual things are not Abstract. Not Composite abstract.
56-33 (the human 'Charlie Chaplin')	- Composite tangible concrete thing	**YES**	Conceptual things are not Concrete. Not Composite tangible concrete.
56-33 (the human 'Charlie Chaplin')	- Composite intangible concrete thing	**YES**	Conceptual things are not Concrete. Not Composite intangible concrete.
56-33 (the human 'Charlie Chaplin')	- Elemental concrete thing	**YES**	Conceptual things are not Concrete. Not Elemental concrete.
56-33 (the human 'Charlie Chaplin')	- Elemental abstract thing	**YES**	Conceptual things are not Abstract. Not Elemental abstract.
56-33 (the human 'Charlie Chaplin')	- Elemental tangible concrete thing	**YES**	Conceptual things are not Concrete. Not Elemental tangible concrete.
56-33 (the human 'Charlie Chaplin')	- Elemental intangible concrete thing	**YES**	Conceptual things are not Concrete. Not Elemental intangible concrete.
56-33 (the human 'Charlie Chaplin')	- Elemental conceptual thing	**YES**	The Conceptual thing is Composite (C19=- #). Disjoint & complete: Composite / Elemental. Not Elemental conceptual.
56-33 (the human 'Charlie Chaplin')	- Elemental noetic thing	**YES**	The Conceptual thing is Composite (C19=- #). Disjoint & complete: Composite / Elemental. Not Elemental noetic.
59-04 <a date>	54-02 can represent a Thing	**YES**	Abstract things do not represent other Things — they have no referent relationship (58-03). Representation is a Conceptual-thing characteristic.
59-04 <a date>	54-02 in reality	**YES**	Abstract things exist in the mind, not in reality (reasoning C: you observe 387 dogs, not 387 itself). Disjoint & complete: in a mind / in reality.
59-04 <a date>	55-12 collection of Things	**YES**	Abstract things are Elemental (not collections of component Things). A number is not decomposed into sub-Things within EO. C20 (Elemental) is marked - # for Abstract things, confirming this.
59-04 <a date>	56-29 physical dimensions	**YES**	Abstract things are purely mental constructs with no physical dimensions whatsoever.
59-04 <a date>	56-29 conceived	**YES**	"Conceived" (C10) refers to intangible Concrete things existing in a conceived world (56-29). Abstract things exist in the mind differently — they have no referent, and their being-in-the-mind is captured by C00, not C10.
59-04 <a date>	56-19 Concrete thing	**YES**	Disjoint: Concrete thing, Abstract thing. A number/boolean/date is Abstract.
59-04 <a date>	56-29 Tangible concrete thing	**YES**	Abstract things are not Concrete at all; hence not Tangible concrete.

Item	Absent Characteristic	Correctly Absent	Justification
59-04 <a date>	56-29 Intangible concrete thing	**YES**	Abstract things are not Concrete at all; hence not Intangible concrete.
59-04 <a date>	56-33 Conceptual thing	**YES**	Abstract things are not Conceptual things. These are distinct categories in EO.
59-04 <a date>	66-05 Composite thing	**YES**	Abstract things are Elemental (not collections of sub-Things). C20 (Elemental thing) is marked - # for Abstract things, and Composite/Elemental are disjoint & complete.
59-04 <a date>	- Composite concrete thing	**YES**	Abstract things are not Concrete; hence not Composite concrete.
59-04 <a date>	- Composite abstract thing	**YES**	Abstract things are Elemental (C20=-#). Disjoint & complete: Composite / Elemental. Not Composite abstract.
59-04 <a date>	- Composite tangible concrete thing	**YES**	Abstract things are not Concrete. Not Composite tangible concrete.
59-04 <a date>	- Composite intangible concrete thing	**YES**	Abstract things are not Concrete. Not Composite intangible concrete.
59-04 <a date>	- Composite conceptual thing	**YES**	Abstract things are not Conceptual. Not Composite conceptual.
59-04 <a date>	- Composite noetic thing	**YES**	Abstract things are Elemental (C20=-#). Disjoint & complete: Composite / Elemental. Not Composite noetic (though Noetic and Elemental).
59-04 <a date>	- Elemental concrete thing	**YES**	Abstract things are not Concrete. Not Elemental concrete.
59-04 <a date>	- Elemental tangible concrete thing	**YES**	Abstract things are not Concrete. Not Elemental tangible concrete.
59-04 <a date>	- Elemental intangible concrete thing	**YES**	Abstract things are not Concrete. Not Elemental intangible concrete.
59-04 <a date>	- Elemental conceptual thing	**YES**	Abstract things are not Conceptual. Not Elemental conceptual.
59-08 [an insurance policy]	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
59-08 [an insurance policy]	54-02 can represent a Thing	**YES**	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
59-08 [an insurance policy]	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
59-08 [an insurance policy]	56-29 physical dimensions	**YES**	Intangible concrete things have no physical dimensions; they exist in a conceived world (56-29). Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
59-08 [an insurance policy]	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.

Item	Absent Characteristic	Correctly Absent	Justification
59-08 [an insurance policy]	56-20 Abstract thing	YES	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
59-08 [an insurance policy]	56-29 Tangible concrete thing	YES	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Intangible concrete things are not Tangible.
59-08 [an insurance policy]	56-33 Conceptual thing	YES	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
59-08 [an insurance policy]	- Elemental thing	YES	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
59-08 [an insurance policy]	- Noetic thing	YES	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
59-08 [an insurance policy]	- Composite abstract thing	YES	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
59-08 [an insurance policy]	- Composite tangible concrete thing	YES	The item is Intangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite tangible concrete.
59-08 [an insurance policy]	- Composite conceptual thing	YES	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
59-08 [an insurance policy]	- Composite noetic thing	YES	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
59-08 [an insurance policy]	- Elemental concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
59-08 [an insurance policy]	- Elemental abstract thing	YES	Concrete things are not Abstract. Not Elemental abstract.
59-08 [an insurance policy]	- Elemental tangible concrete thing	YES	The item is Intangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental tangible concrete.
59-08 [an insurance policy]	- Elemental intangible concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental intangible concrete.
59-08 [an insurance policy]	- Elemental conceptual thing	YES	Concrete things are not Conceptual. Not Elemental conceptual.
59-08 [an insurance policy]	- Elemental noetic thing	YES	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
59-08 [an employment contract]	54-01 in a mind	YES	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
59-08 [an employment contract]	54-02 can represent a Thing	YES	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
59-08 [an employment contract]	- single Thing	YES	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are

Item	Absent Characteristic	Correctly Absent	Justification
			either collections or have no collection role).
59-08 [an employment contract]	56-29 physical dimensions	**YES**	Intangible concrete things have no physical dimensions; they exist in a conceived world (56-29). Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
59-08 [an employment contract]	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
59-08 [an employment contract]	56-20 Abstract thing	**YES**	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
59-08 [an employment contract]	56-29 Tangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Intangible concrete things are not Tangible.
59-08 [an employment contract]	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
59-08 [an employment contract]	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
59-08 [an employment contract]	- Noetic thing	**YES**	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
59-08 [an employment contract]	- Composite abstract thing	**YES**	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
59-08 [an employment contract]	- Composite tangible concrete thing	**YES**	The item is Intangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite tangible concrete.
59-08 [an employment contract]	- Composite conceptual thing	**YES**	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
59-08 [an employment contract]	- Composite noetic thing	**YES**	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
59-08 [an employment contract]	- Elemental concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
59-08 [an employment contract]	- Elemental abstract thing	**YES**	Concrete things are not Abstract. Not Elemental abstract.
59-08 [an employment contract]	- Elemental tangible concrete thing	**YES**	The item is Intangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental tangible concrete.
59-08 [an employment contract]	- Elemental intangible concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental intangible concrete.
59-08 [an employment contract]	- Elemental conceptual thing	**YES**	Concrete things are not Conceptual. Not Elemental conceptual.
59-08 [an employment contract]	- Elemental noetic thing	**YES**	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.

Item	Absent Characteristic	Correctly Absent	Justification
59-08 [a car]	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
59-08 [a car]	54-02 can represent a Thing	**YES**	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
59-08 [a car]	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
59-08 [a car]	56-29 conceived	**YES**	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
59-08 [a car]	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
59-08 [a car]	56-20 Abstract thing	**YES**	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
59-08 [a car]	56-29 Intangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
59-08 [a car]	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
59-08 [a car]	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
59-08 [a car]	- Noetic thing	**YES**	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
59-08 [a car]	- Composite abstract thing	**YES**	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
59-08 [a car]	- Composite intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
59-08 [a car]	- Composite conceptual thing	**YES**	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
59-08 [a car]	- Composite noetic thing	**YES**	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
59-08 [a car]	- Elemental concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
59-08 [a car]	- Elemental abstract thing	**YES**	Concrete things are not Abstract. Not Elemental abstract.
59-08 [a car]	- Elemental tangible concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete:

Item	Absent Characteristic	Correctly Absent	Justification
			Composite / Elemental. Not Elemental tangible concrete.
59-08 [a car]	- Elemental intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
59-08 [a car]	- Elemental conceptual thing	**YES**	Concrete things are not Conceptual. Not Elemental conceptual.
59-08 [a car]	- Elemental noetic thing	**YES**	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
59-08 [a house]	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
59-08 [a house]	54-02 can represent a Thing	**YES**	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
59-08 [a house]	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
59-08 [a house]	56-29 conceived	**YES**	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
59-08 [a house]	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
59-08 [a house]	56-20 Abstract thing	**YES**	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
59-08 [a house]	56-29 Intangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
59-08 [a house]	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
59-08 [a house]	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
59-08 [a house]	- Noetic thing	**YES**	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
59-08 [a house]	- Composite abstract thing	**YES**	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
59-08 [a house]	- Composite intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
59-08 [a house]	- Composite conceptual thing	**YES**	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.

Item	Absent Characteristic	Correctly Absent	Justification
59-08 [a house]	- Composite noetic thing	**YES**	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
59-08 [a house]	- Elemental concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
59-08 [a house]	- Elemental abstract thing	**YES**	Concrete things are not Abstract. Not Elemental abstract.
59-08 [a house]	- Elemental tangible concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental tangible concrete.
59-08 [a house]	- Elemental intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
59-08 [a house]	- Elemental conceptual thing	**YES**	Concrete things are not Conceptual. Not Elemental conceptual.
59-08 [a house]	- Elemental noetic thing	**YES**	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
62-21 [a student]	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
62-21 [a student]	54-02 can represent a Thing	**YES**	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
62-21 [a student]	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
62-21 [a student]	56-29 conceived	**YES**	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
62-21 [a student]	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
62-21 [a student]	56-20 Abstract thing	**YES**	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
62-21 [a student]	56-29 Intangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
62-21 [a student]	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
62-21 [a student]	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.

Item	Absent Characteristic	Correctly Absent	Justification
62-21 [a student]	- Noetic thing	**YES**	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
62-21 [a student]	- Composite abstract thing	**YES**	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
62-21 [a student]	- Composite intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
62-21 [a student]	- Composite conceptual thing	**YES**	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
62-21 [a student]	- Composite noetic thing	**YES**	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
62-21 [a student]	- Elemental concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
62-21 [a student]	- Elemental abstract thing	**YES**	Concrete things are not Abstract. Not Elemental abstract.
62-21 [a student]	- Elemental tangible concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental tangible concrete.
62-21 [a student]	- Elemental intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
62-21 [a student]	- Elemental conceptual thing	**YES**	Concrete things are not Conceptual. Not Elemental conceptual.
62-21 [a student]	- Elemental noetic thing	**YES**	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
65-02 (a chair)	54-02 in reality	**YES**	Conceptual things are in the mind (59-10). Disjoint & complete: in a mind / in reality. A Conceptual thing is not in reality.
65-02 (a chair)	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
65-02 (a chair)	56-29 physical dimensions	**YES**	Conceptual things are in the mind; they have no physical dimensions.
65-02 (a chair)	56-29 conceived	**YES**	"Conceived" (C10) denotes intangible Concrete things in a conceived world. Conceptual things are in the mind (C00=- #), which is a different characteristic from the intangible-concrete "conceived world."
65-02 (a chair)	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is specific to Abstract things (58-03). Conceptual things actively DO represent Things (56-21) — the opposite holds.
65-02 (a chair)	56-19 Concrete thing	**YES**	Disjoint: Concrete thing, Conceptual thing. A Conceptual thing is not Concrete.

Item	Absent Characteristic	Correctly Absent	Justification
65-02 (a chair)	56-20 Abstract thing	YES	Conceptual things are not Abstract things. These are distinct categories in EO.
65-02 (a chair)	56-29 Tangible concrete thing	YES	Conceptual things are not Concrete; hence not Tangible concrete.
65-02 (a chair)	56-29 Intangible concrete thing	YES	Conceptual things are not Concrete; hence not Intangible concrete.
65-02 (a chair)	- Elemental thing	YES	The Conceptual thing is marked as Composite (C19=- #). Disjoint & complete: Composite / Elemental.
65-02 (a chair)	- Composite concrete thing	YES	Conceptual things are not Concrete. Disjoint: Concrete / Conceptual. Not Composite concrete.
65-02 (a chair)	- Composite abstract thing	YES	Conceptual things are not Abstract. Not Composite abstract.
65-02 (a chair)	- Composite tangible concrete thing	YES	Conceptual things are not Concrete. Not Composite tangible concrete.
65-02 (a chair)	- Composite intangible concrete thing	YES	Conceptual things are not Concrete. Not Composite intangible concrete.
65-02 (a chair)	- Elemental concrete thing	YES	Conceptual things are not Concrete. Not Elemental concrete.
65-02 (a chair)	- Elemental abstract thing	YES	Conceptual things are not Abstract. Not Elemental abstract.
65-02 (a chair)	- Elemental tangible concrete thing	YES	Conceptual things are not Concrete. Not Elemental tangible concrete.
65-02 (a chair)	- Elemental intangible concrete thing	YES	Conceptual things are not Concrete. Not Elemental intangible concrete.
65-02 (a chair)	- Elemental conceptual thing	YES	The Conceptual thing is Composite (C19=- #). Disjoint & complete: Composite / Elemental. Not Elemental conceptual.
65-02 (a chair)	- Elemental noetic thing	YES	The Conceptual thing is Composite (C19=- #). Disjoint & complete: Composite / Elemental. Not Elemental noetic.
66-05 [a ship]	54-01 in a mind	YES	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
66-05 [a ship]	54-02 can represent a Thing	YES	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
66-05 [a ship]	- single Thing	YES	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
66-05 [a ship]	56-29 conceived	YES	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
66-05 [a ship]	58-03 cannot represent a Thing	YES	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.

Item	Absent Characteristic	Correctly Absent	Justification
66-05 [a ship]	56-20 Abstract thing	**YES**	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
66-05 [a ship]	56-29 Intangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
66-05 [a ship]	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
66-05 [a ship]	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
66-05 [a ship]	- Noetic thing	**YES**	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
66-05 [a ship]	- Composite abstract thing	**YES**	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
66-05 [a ship]	- Composite intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
66-05 [a ship]	- Composite conceptual thing	**YES**	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
66-05 [a ship]	- Composite noetic thing	**YES**	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
66-05 [a ship]	- Elemental concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
66-05 [a ship]	- Elemental abstract thing	**YES**	Concrete things are not Abstract. Not Elemental abstract.
66-05 [a ship]	- Elemental tangible concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental tangible concrete.
66-05 [a ship]	- Elemental intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
66-05 [a ship]	- Elemental conceptual thing	**YES**	Concrete things are not Conceptual. Not Elemental conceptual.
66-05 [a ship]	- Elemental noetic thing	**YES**	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
66-05 [a plank]	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
66-05 [a plank]	54-02 can represent a Thing	**YES**	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
66-05 [a plank]	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are

Item	Absent Characteristic	Correctly Absent	Justification
			either collections or have no collection role).
66-05 [a plank]	56-29 conceived	YES	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
66-05 [a plank]	58-03 cannot represent a Thing	YES	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
66-05 [a plank]	56-20 Abstract thing	YES	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
66-05 [a plank]	56-29 Intangible concrete thing	YES	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
66-05 [a plank]	56-33 Conceptual thing	YES	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
66-05 [a plank]	- Elemental thing	YES	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
66-05 [a plank]	- Noetic thing	YES	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
66-05 [a plank]	- Composite abstract thing	YES	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
66-05 [a plank]	- Composite intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
66-05 [a plank]	- Composite conceptual thing	YES	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
66-05 [a plank]	- Composite noetic thing	YES	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
66-05 [a plank]	- Elemental concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
66-05 [a plank]	- Elemental abstract thing	YES	Concrete things are not Abstract. Not Elemental abstract.
66-05 [a plank]	- Elemental tangible concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental tangible concrete.
66-05 [a plank]	- Elemental intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
66-05 [a plank]	- Elemental conceptual thing	YES	Concrete things are not Conceptual. Not Elemental conceptual.
66-05 [a plank]	- Elemental noetic thing	YES	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.

Item	Absent Characteristic	Correctly Absent	Justification
66-14 [a butterfly]	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
66-14 [a butterfly]	54-02 can represent a Thing	**YES**	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
66-14 [a butterfly]	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
66-14 [a butterfly]	56-29 conceived	**YES**	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
66-14 [a butterfly]	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
66-14 [a butterfly]	56-20 Abstract thing	**YES**	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
66-14 [a butterfly]	56-29 Intangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
66-14 [a butterfly]	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
66-14 [a butterfly]	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
66-14 [a butterfly]	- Noetic thing	**YES**	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
66-14 [a butterfly]	- Composite abstract thing	**YES**	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
66-14 [a butterfly]	- Composite intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
66-14 [a butterfly]	- Composite conceptual thing	**YES**	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
66-14 [a butterfly]	- Composite noetic thing	**YES**	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
66-14 [a butterfly]	- Elemental concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
66-14 [a butterfly]	- Elemental abstract thing	**YES**	Concrete things are not Abstract. Not Elemental abstract.
66-14 [a butterfly]	- Elemental tangible concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete:

Item	Absent Characteristic	Correctly Absent	Justification
			Composite / Elemental. Not Elemental tangible concrete.
66-14 [a butterfly]	- Elemental intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
66-14 [a butterfly]	- Elemental conceptual thing	**YES**	Concrete things are not Conceptual. Not Elemental conceptual.
66-14 [a butterfly]	- Elemental noetic thing	**YES**	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
66-14 [a caterpillar]	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
66-14 [a caterpillar]	54-02 can represent a Thing	**YES**	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
66-14 [a caterpillar]	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
66-14 [a caterpillar]	56-29 conceived	**YES**	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
66-14 [a caterpillar]	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
66-14 [a caterpillar]	56-20 Abstract thing	**YES**	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
66-14 [a caterpillar]	56-29 Intangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
66-14 [a caterpillar]	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
66-14 [a caterpillar]	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
66-14 [a caterpillar]	- Noetic thing	**YES**	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
66-14 [a caterpillar]	- Composite abstract thing	**YES**	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
66-14 [a caterpillar]	- Composite intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
66-14 [a caterpillar]	- Composite conceptual thing	**YES**	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.

Item	Absent Characteristic	Correctly Absent	Justification
66-14 [a caterpillar]	- Composite noetic thing	**YES**	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
66-14 [a caterpillar]	- Elemental concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
66-14 [a caterpillar]	- Elemental abstract thing	**YES**	Concrete things are not Abstract. Not Elemental abstract.
66-14 [a caterpillar]	- Elemental tangible concrete thing	**YES**	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental tangible concrete.
66-14 [a caterpillar]	- Elemental intangible concrete thing	**YES**	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
66-14 [a caterpillar]	- Elemental conceptual thing	**YES**	Concrete things are not Conceptual. Not Elemental conceptual.
66-14 [a caterpillar]	- Elemental noetic thing	**YES**	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
66-25 [a brain]	54-01 in a mind	**YES**	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
66-25 [a brain]	54-02 can represent a Thing	**YES**	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
66-25 [a brain]	- single Thing	**YES**	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
66-25 [a brain]	56-29 conceived	**YES**	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
66-25 [a brain]	58-03 cannot represent a Thing	**YES**	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
66-25 [a brain]	56-20 Abstract thing	**YES**	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.
66-25 [a brain]	56-29 Intangible concrete thing	**YES**	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
66-25 [a brain]	56-33 Conceptual thing	**YES**	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
66-25 [a brain]	- Elemental thing	**YES**	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.

Item	Absent Characteristic	Correctly Absent	Justification
66-25 [a brain]	- Noetic thing	YES	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
66-25 [a brain]	- Composite abstract thing	YES	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
66-25 [a brain]	- Composite intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
66-25 [a brain]	- Composite conceptual thing	YES	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
66-25 [a brain]	- Composite noetic thing	YES	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
66-25 [a brain]	- Elemental concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
66-25 [a brain]	- Elemental abstract thing	YES	Concrete things are not Abstract. Not Elemental abstract.
66-25 [a brain]	- Elemental tangible concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental tangible concrete.
66-25 [a brain]	- Elemental intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
66-25 [a brain]	- Elemental conceptual thing	YES	Concrete things are not Conceptual. Not Elemental conceptual.
66-25 [a brain]	- Elemental noetic thing	YES	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.
66-26 [a motor block]	54-01 in a mind	YES	Concrete things are in reality (54-02, 59-08), not in a mind. Disjoint & complete: in a mind / in reality.
66-26 [a motor block]	54-02 can represent a Thing	YES	Representing Things is a characteristic of Conceptual things (56-21), not of Concrete things. Concrete things are the referents, not the representers.
66-26 [a motor block]	- single Thing	YES	- single Thing is a placeholder complementary to collection of Things. It denotes a single member, not a set. No item in this formal context is a single Thing in the sense of C05 (all items are either collections or have no collection role).
66-26 [a motor block]	56-29 conceived	YES	Tangible concrete things are physically observable (56-29); they are not "conceived" in the intangible sense. Disjoint & complete: Tangible concrete thing / Intangible concrete thing.
66-26 [a motor block]	58-03 cannot represent a Thing	YES	"Cannot represent a Thing" is a characteristic specific to Abstract things (58-03). Concrete things are not characterised this way; they simply have no representational role in EO.
66-26 [a motor block]	56-20 Abstract thing	YES	Disjoint: Concrete thing, Abstract thing. All Concrete items are not Abstract.

Item	Absent Characteristic	Correctly Absent	Justification
66-26 [a motor block]	56-29 Intangible concrete thing	YES	Disjoint & complete: Tangible concrete thing / Intangible concrete thing. Tangible concrete things are not Intangible.
66-26 [a motor block]	56-33 Conceptual thing	YES	Disjoint: Concrete thing, Conceptual thing. All Concrete items are not Conceptual.
66-26 [a motor block]	- Elemental thing	YES	The item is marked as Composite (C19=- # or explicit X). Disjoint & complete: Composite thing / Elemental thing.
66-26 [a motor block]	- Noetic thing	YES	Concrete things exist in reality (54-02, 59-08). Noetic things exist in the mind. Concrete things are not Noetic.
66-26 [a motor block]	- Composite abstract thing	YES	Concrete things are not Abstract. Disjoint: Concrete / Abstract. Not Composite abstract.
66-26 [a motor block]	- Composite intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Composite intangible concrete.
66-26 [a motor block]	- Composite conceptual thing	YES	Concrete things are not Conceptual. Disjoint: Concrete / Conceptual. Not Composite conceptual.
66-26 [a motor block]	- Composite noetic thing	YES	Concrete things exist in reality, not in the mind. Not Noetic; not Composite noetic.
66-26 [a motor block]	- Elemental concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental concrete.
66-26 [a motor block]	- Elemental abstract thing	YES	Concrete things are not Abstract. Not Elemental abstract.
66-26 [a motor block]	- Elemental tangible concrete thing	YES	The item is Composite (C19=- # or explicit). Disjoint & complete: Composite / Elemental. Not Elemental tangible concrete.
66-26 [a motor block]	- Elemental intangible concrete thing	YES	The item is Tangible concrete. Disjoint & complete: Tangible / Intangible. Not Elemental intangible concrete.
66-26 [a motor block]	- Elemental conceptual thing	YES	Concrete things are not Conceptual. Not Elemental conceptual.
66-26 [a motor block]	- Elemental noetic thing	YES	Concrete things are not Noetic (they exist in reality, not the mind). Not Elemental noetic.

7. Check 6 — ALL Sentence Interpretations

All 96 discourse sentences are assessed. Rows marked UNCHANGED have an interpretation identical to the original sentence and are trivially correct. Rows with modifications are assessed for consistency with the formal context and EO terminology.

Key	Original Sentence	Interpretation	Changed	Result	Assessment
54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.	YES	CORRECT	Narrows "Thoughts are things" to Conceptual things. Correct EO refinement.

Key	Original Sentence	Interpretation	Changed	Result	Assessment
54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things in reality), called referents in the semiotic triangle.	YES	**CORRECT**	"Things in reality" is synonymous with "outside the mind." EO terms capitalised correctly.
54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to deno…	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to deno…	NO	**CORRECT**	Interpretation equals sentence. No modification needed; the sentence applies directly.
54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.	YES	**CORRECT**	"Things" capitalised; otherwise identical. Capitalisation is the EO convention for the concept.
56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a World consists of distinguishable Things.	YES	**CORRECT**	"World" and "Things" capitalised; "concrete" dropped. Dropping "concrete" is consistent with the formal context treating all distinguishable Things in a world, per 56-01 itself.
56-02	A Thing is said to be an Object with an assigned Form of Features.	A Thing is said to be an Object with an assigned Form of Features.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
56-03	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
56-05	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
56-09	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
56-12	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
56-13	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object. Bunge names a…	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object. Bunge names a…	NO	**CORRECT**	Interpretation equals sentence. No modification needed.

Key	Original Sentence	Interpretation	Changed	Result	Assessment
56-14	In this bare Object resides the identity of the thing.	In this bare Object resides the identity of the thing.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
56-16	Only when a Form has at least one Feature, there is a Thing.	Only when a Form has at least one Feature, there is a Thing.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
56-19	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
56-20	Abstract objects are Objects of Abstract things, like numbers and boolean values.	Abstract objects are Objects of Abstract things, like numbers and boolean values.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
56-21	When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This impli…	When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates t…	YES	**CORRECT**	"Observing" replaced by "recognising" throughout — consistent EO refinement used across B_discourse.
56-24	It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.	It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.	YES	**CORRECT**	"Observe" → "recognise"; "classes" → "Classes". Consistent EO term capitalisation.
56-27	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals.	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we recognise, and then conclude that the Things belong to certain Classes, like humans and rentals.	YES	**CORRECT**	"Observe" → "recognise". Same pattern as 56-21 and 56-24.
56-28	Consequently, Conceptual things are Conceptual objects with assigned Form, a collection of Features.	Consequently, Conceptual things are Conceptual objects with assigned Form, a collection of Features.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
56-29	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that o…	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to Concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that o…	YES	**CORRECT**	"Concrete things" → "Concrete things"; "conceived world" → "Conceived world". Capitalisation of proper EO terms.
56-30	The example referent in Fig. 5.3 (left) is a concrete thing of the type human.	The example referent in Fig. 5.3 (left) is a Concrete thing of the Type human.	YES	**CORRECT**	"Concrete thing" and "Type" capitalised. Correct EO usage.
56-32	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
56-33	To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin').	To avoid confusion, we will refer to the Concrete thing as [the human 'Charlie Chaplin'], and to the corresponding Conceptual thing as (the human 'Charlie Chaplin').	YES	**CORRECT**	<...> brackets for concrete thing replaced by [...] per bracket convention (Constraint 8). Correct substitution.
56-34	For Concrete objects and classes, we use the bracket pair "[" and "]".	For Concrete objects and classes, we use the bracket pair "[" and "]".	NO	**CORRECT**	"classes" → "Classes". Correct capitalisation.

Key	Original Sentence	Interpretation	Changed	Result	Assessment
57-01	Fig. 5.3 (right) shows an example of applying the semiotic triangle to classes.	Fig. 5.3 (right) shows an example of applying the semiotic triangle to classes.	NO	**CORRECT**	"classes" → "Classes". Correct capitalisation.
57-02	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
57-03	It is represented in our mind by the thought (the class HUMAN).	The Concrete things of Class [HUMAN] are represented by Conceptual things from Class (HUMAN).	YES	**CORRECT**	Interpretation extends class-level statement to individual Thing level: Concrete things of [HUMAN] represented by Conceptual things of (HUMAN). Consistent with 60-25 and the formal context.
57-04	The string "HUMAN" is a name from the name class for classes.	The string "HUMAN" is a name from the name class for classes.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
57-06	The referents we consider now are thus the Concrete objects of Concrete things.	The referents we consider now are thus the Concrete objects of Concrete things.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
57-09	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
58-02	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
58-03	As indicated by the question mark, Abstract things don't represent any Concrete things.	As indicated by the question mark, Abstract things don't represent any Concrete things.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
58-04	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
58-05	Well-known abstract things are numbers.	Well-known abstract things are numbers.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
58-06	The name "the number '387'" signifies the abstract thing (the number '387').	The name "the number '387'" signifies the abstract thing (the number '387').	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
58-07	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
58-08	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
58-14	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.

Key	Original Sentence	Interpretation	Changed	Result	Assessment
59-01	If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity.	If the Thing of a Unary fact is a Concrete thing, the Concrete unary fact is commonly called an Entity.	YES	**CORRECT**	Correctly identifies that it is the Concrete unary fact (not the Thing itself) that is called an Entity. Precise ontological refinement.
59-02	As an example, one commonly speaks of the entity Charlie Chaplin, thereby referring to the concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".	As an example, one commonly speaks of the Entity Charlie Chaplin, thereby referring to the Concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".	YES	**CORRECT**	"Entity" and "Concrete unary fact" capitalised. Consistent with the interpretation of 59-01.
59-03	Likewise, if the thing of a unary fact is Abstract, the Abstract thing is commonly called a Value.	Likewise, if the Thing of a Unary fact is Abstract, the Abstract unary fact is commonly called a Value.	YES	**CORRECT**	Analogous to 59-01: Abstract unary fact called a Value. Correct.
59-04	As an example, day 2411108 is a value on the time scale Julian Date.	As an example, date 2411108 is a value on the time scale Julian Date.	YES	**CORRECT**	"day" → "date" aligns with formal context item <a date>. Correct minor refinement.
59-07	In answering these questions, we will adhere strictly to the distinctions we have made between concrete and conceptual things.	In answering these questions, we will adhere strictly to the distinctions we have made between Concrete and Conceptual things.	YES	**CORRECT**	"concrete and conceptual" → "Concrete and Conceptual". Capitalisation only.
59-08	Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy fo…	Concrete things are the Things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy fo…	YES	**CORRECT**	"things" → "Things" (EO capitalisation). Otherwise identical.
59-09	As said, concrete things can be tangible (like a house) and intangible (like a rental).	As said, concrete things can be tangible (like a house) and intangible (like a rental).	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
59-10	Conceptual things are things in the mind.	Conceptual things are Things in the mind.	YES	**CORRECT**	"things" → "Things". EO capitalisation.
60-05	A Type is defined as a template that the mind applies when we consider Concrete things (both tangible and intangible).	A Type is defined as a template that the mind applies when we consider Concrete things (both tangible and intangible).	YES	**CORRECT**	Interpretation equals sentence. No modification needed.
60-06	The Form of a concrete thing is the collection of its Features.	The Form of a concrete thing is the collection of its Features.	NO	**CORRECT**	"concrete thing" → "Concrete thing". EO capitalisation.
60-08	In applying a type, the mind puts, so to speak, the type's template over the things in the world, and sieves out the ones that conform.	In applying a type, the mind puts, so to speak, the type's template over the Things in the World, and sieves out the ones that conform.	YES	**CORRECT**	"things in the world" → "Things in the World". EO capitalisation.
60-10	Let us next assume that there is a Concrete thing whose Concrete object we signify by [object 701], and let us study the Concrete thing that is indicated as [some thing].	Let us next assume that there is a Concrete thing whose Concrete object we signify by [object 701], and let us study the Concrete thing that is indicated as [some thing].	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
60-11	To illustrate that Things are Objects with assigned Form, the symbol for the Thing in the semiotic mill is a magenta coloured disk with protrusions,	To illustrate that Things are Objects with assigned Form, the symbol for the Thing in the semiotic mill is a magenta coloured disk with protrusions,	NO	**CORRECT**	Interpretation equals sentence. No modification needed.

Key	Original Sentence	Interpretation	Changed	Result	Assessment
	where every protrusion stands for a Feature.	where every protrusion stands for a Feature.			
60-13	When putting the template of the type human over the thing, we conclude that it conforms to the type human, because one of its protrusions fits the protrusion of the type human.	When putting the template of the Type human over the Thing, we conclude that it conforms to the Type human, because one of its protrusions fits the protrusion of the Type human.	YES	**CORRECT**	"type" → "Type", "thing" → "Thing". EO capitalisation.
60-20	The Conceptual things of facts of the same type, constitute a Conceptual class of which each of the Conceptual things is a member.	The Conceptual things of facts of the same type, constitute a Conceptual class of which each of the Conceptual things is a member.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
60-21	Let us call the Conceptual class of Things that are predicated by Facts of the Type human, HUMAN.	Let us call the Conceptual class of Conceptual things that are predicated by Facts of the Type human, (HUMAN).	YES	**CORRECT**	"Things" → "Conceptual things"; "HUMAN" → "(HUMAN)". Applies bracket convention to Conceptual class. Correct.
60-22	This class is called the extension of the type human.	This Conceptual class is called the extension of the Type human.	YES	**CORRECT**	"class" → "Conceptual class"; "type" → "Type". EO capitalisation and precision.
60-23	Conversely, the type human is called the intension of the class HUMAN.	Conversely, the type human is called the intension of the class HUMAN.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
60-25	Lastly, the conceptual class (HUMAN) represents the concrete class [HUMAN].	Lastly, the Conceptual things of the Conceptual class (HUMAN) represents the Concrete things of Concrete class [HUMAN].	YES	**CORRECT**	Extends class-level representation to individual Thing level — consistent with formal context treatment.
61-04	A Fact starts to exist either as soon as the Form of a thing conforms to a Type, or as soon as a Type starts to exist to which the Thing conforms.	A Fact starts to exist either as soon as the Thing conforms to a Type, or as soon as a Type starts to exist to which the Thing conforms.	YES	**CORRECT**	"Form of a thing conforms to a Type" simplified to "Thing conforms to a Type" — standard EO convention.
61-06	Through this event, a new entity of the type butterfly starts to exist.	Through this event, a new Entity of the Type butterfly starts to exist.	YES	**CORRECT**	"entity" → "Entity", "type" → "Type". EO capitalisation.
61-09	A Fact ends to exist either as soon as the Form of the Thing that the Fact is about, does not conform anymore to the Type, or as soon as a Type to which it conforms, ends to exist.	A Fact ends to exist either as soon as the Thing that the Fact is about, does not conform anymore to the Type, or as soon as a Type to which it conforms, ends to exist.	YES	**CORRECT**	Same simplification as 61-04. Correct.
61-11	After this event has occurred, the thing does not conform anymore to the type caterpillar.	After this event has occurred, the Thing does not conform anymore to the Type caterpillar.	YES	**CORRECT**	"thing" → "Thing", "type" → "Type". EO capitalisation.
61-12	Consequently, the entity of the type caterpillar ends to exist.	Consequently, the Entity of the Type caterpillar ends to exist.	YES	**CORRECT**	"entity" → "Entity", "type" → "Type". EO capitalisation.
61-16	As Mario Bunge puts it, a world is not a collection of facts, pace Wittgenstein, but a collection of things.	As Mario Bunge puts it, a World is not a collection of Facts, pace Wittgenstein, but a collection of Things.	YES	**CORRECT**	"world" → "World", "facts" → "Facts", "things" → "Things". EO capitalisation.
61-17	The knowledge of these things, and thus one's factual knowledge of the world, consists of conceptual facts.	The knowledge of these Things, and thus one's factual knowledge of the world, consists of Conceptual facts.	YES	**CORRECT**	"things" → "Things"; "conceptual facts" → "Conceptual facts". EO capitalisation.

Key	Original Sentence	Interpretation	Changed	Result	Assessment
62-09	As is explained in Chap. 8, the entity types that occur in product kinds are proper candidates for being declared.	As is explained in Chap. 8, the Entity types that occur in Product kinds are proper candidates for being declared.	YES	**CORRECT**	"entity types" → "Entity types"; "product kinds" → "Product kinds". EO capitalisation.
62-21	When this is not the case anymore, the student entity ceases to exist.	When this is not the case anymore, the student Entity ceases to exist.	YES	**CORRECT**	"entity" → "Entity". EO capitalisation.
62-22	By generalisation is understood that one defines a new type as the union of the extensions, so the classes, of two or more existing types (which may be declared or derived).	By generalisation is understood that one defines a new Type as the union of the extensions, so the Classes, of two or more existing types (which may be declared or derived).	YES	**CORRECT**	"type" → "Type"; "classes" → "Classes". EO capitalisation.
62-24	We can then define the type vehicle as the intension of the class (cf. Fig. 5.7) that equals the set-theoretic union of the extensions of car, boat and aircraft.	We can then define the Type vehicle as the intension of the Class (cf. Fig. 5.7) that equals the set-theoretic union of the extensions of car, boat and aircraft.	YES	**CORRECT**	"type" → "Type"; "class" → "Class". EO capitalisation.
64-01	The individuals in all these classes are identified by the concrete identity they have got in the class genus.	The Things in all these Classes are identified by the Things in class genus.	YES	**CORRECT**	"individuals" → "Things"; "classes" → "Classes"; second "concrete identity" clause restated as "Things in class genus". Correct EO vocabulary.
64-11	The type human, as discussed, is typically a constructional type: it is based exclusively on the properties of things.	The Type human, as discussed, is typically a constructional Type: it is based exclusively on the Form of a Thing.	YES	**CORRECT**	"type" → "Type"; "properties of things" → "Form of a Thing". In EO, properties are Features of Form — correct substitution.
65-01	Figure 5.9 exhibits the conceptualisation of things of the type sit-on-able.	Figure 5.9 exhibits the conceptualisation of Things of the Type sit-on-able.	YES	**CORRECT**	"things" → "Things"; "type" → "Type". EO capitalisation.
65-02	It shows the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair). Relationships are detailed in the TAO theory.	It shows the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair). Relationships are detailed in the TAO theory.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
65-11	The type 'chair model Chaplin' can be specified precisely enough from the construction perspective, in order to decide whether a thing conforms to the type or not.	The Type 'chair model Chaplin' can be specified precisely enough from the construction perspective, in order to decide whether a Thing conforms to the Type or not.	YES	**CORRECT**	"type" → "Type"; "thing" → "Thing". EO capitalisation.
66-03	Its general questions are: When does a thing change? and: If a thing changes, is it still the same thing (only changed)?	Its general questions are: When does a Thing change? and: If a Thing changes, is it still the same Thing (only changed)?	YES	**CORRECT**	"thing" → "Thing" throughout. EO capitalisation.
66-05	The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their …	The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their …	NO	**CORRECT**	Interpretation equals sentence. No modification needed.

Key	Original Sentence	Interpretation	Changed	Result	Assessment
66-06	In the end, all parts of Theseus' ship are replaced.	In the end, all parts of Theseus' ship are replaced.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
66-08	The key to an appropriate solution of the problem of sameness and change, exemplified by the paradox of Theseus, is to recognise that every Thing has an Object, that is taken as its identity, next to …	The key to an appropriate solution of the problem of sameness and change, exemplified by the paradox of Theseus, is to recognise that every Thing has an Object, that is taken as its identity, next to …	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
66-09	So, even if all the parts of the ship are replaced by other (similar) ones, it is still the same ship.	So, even if all the parts of the ship are replaced by other (similar) ones, it is still the same ship.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
66-10	This argument is supported by the distinction between matter-constant (the ship is composed of the same part instances) and form-constant (the ship is an instance of the type ship), as proposed by Sim…	This argument is supported by the distinction between matter-constant (the ship is composed of the same part instances) and form-constant (the ship is an instance of the Type ship), as proposed by Sim…	YES	**CORRECT**	"type" → "Type". EO capitalisation.
66-11	When the first plank is replaced, the ship is already not the same matter-constant ship anymore.	When the first plank is replaced, the ship is already not the same matter-constant ship anymore.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
66-12	The form-constant argument preserves that it is still a ship of the same type because the whole thing keeps conforming to the type ship (even when all parts are replaced).	The form-constant argument preserves that it is still a ship of the same Type because the whole Thing keeps conforming to the Type ship (even when all parts are replaced).	YES	**CORRECT**	"type" → "Type"; "thing" → "Thing". EO capitalisation.
66-13	However, our argument goes further: Theseus' ship does indeed still conform to the Type ship, but, in addition, it has kept its Object.	However, our argument goes further: Theseus' ship does indeed still conform to the Type ship, but, in addition, it has kept its Object.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
66-14	This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Adorned object after having transformed to being a butterfly from being a caterpillar before.	This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before.	YES	**CORRECT**	"Adorned object" → "Object". "Adorned object" is non-standard; "Object" is the correct EO term. Correct refinement.
66-16	Assume that all parts are replaced and that the 'old' ship is rebuilt by assembling the 'old' parts.	Assume that all parts are replaced and that the 'old' ship is rebuilt by assembling the 'old' parts.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
66-17	Then which ship is Theseus' ship?	Then which ship is Theseus' ship?	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
66-19	We are now confronted with two concrete ships, each of them having its own Concrete object.	We are now confronted with two concrete ships, each of them having its own Concrete object.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
66-20	Most likely, Theseus would consider the current one his ship, not the one that has been re-assembled from the thrown away parts.	Most likely, Theseus would consider the current one his ship, not the one that has been re-assembled from the thrown away parts.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.

Key	Original Sentence	Interpretation	Changed	Result	Assessment
66-24	Many researchers in ontology, like Guarino and Guizzardi, try to solve the problem of sameness and change by assigning the identity of a composite thing to one of its components.	Many researchers in ontology, like Guarino and Guizzardi, try to solve the problem of sameness and change by assigning the identity of a Composite thing to one of its components.	YES	**CORRECT**	"composite thing" → "Composite thing". EO capitalisation.
66-25	Regarding the human body, they consider the human brains as the component that bears the body's identity: as soon as the brains are replaced, it is not the same body anymore.	Regarding the human body, they consider the human brains as the component that bears the body's identity: as soon as the brains are replaced, it is not the same body anymore.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
66-26	With respect to cars, they choose a particular part (mostly the body of the car or the motor block) to bear the car's identity.	With respect to cars, they choose a particular part (mostly the body of the car or the motor block) to bear the car's identity.	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
67-19	The important point to recognise is that a Composite thing has its own Object, independent of the Objects of its components.	The important point to recognise is that a Composite thing has its own Object, independent of the Objects of its components.	YES	**CORRECT**	Interpretation equals sentence. No modification needed.
68-09	The same 'business entity' (upper part) appears to conform to two types, namely to the type purchase (left side), and to the type sale (right side), respectively applied by the buyer and the seller en…	The same 'business entity' (upper part) appears to conform to two Types, namely to the Type purchase (left side), and to the Type sale (right side), respectively applied by the buyer and the seller en…	YES	**CORRECT**	"types" → "Types". EO capitalisation.
68-12	Because the notions of purchase and sale apply to the same business thing, they are called dual notions.	Because the notions of purchase and sale apply to the same Thing, they are called dual notions.	YES	**CORRECT**	"business thing" → "Thing". Correct generalisation to EO term.
69-03	Two names are called synonymous if they signify the same thought (or denote the same referent, cf. Fig. 5).	Two names are called synonymous if they signify the same thought (or denote the same referent, cf. Fig. 5).	NO	**CORRECT**	Interpretation equals sentence. No modification needed.
69-06	Instead they refer to different things, with different properties.	Instead they refer to different Things, with different Features.	YES	**CORRECT**	"properties" → "Features". Correct EO vocabulary (properties = Features in EO).

8. Overall Verdict

Items (Check 1): 23/23 PASS. Characteristics (Check 2): 19/19 PASS. Explicit relationships (Check 3): 73/73 PASS. Inferred relationships (Check 4): 242/242 PASS. Absent relationships (Check 5): 460/460 PASS. Sentence interpretations (Check 6): 96/96 PASS.

VERDICT: FULLY TRACEABLE

The formal context is fully traceable to the EO discourse across all six checks. The extended Check 5 confirms that every absent (— —) cell is correctly absent: in each case the absent characteristic either (i) conflicts with an explicit EO disjoint or partition constraint, (ii) applies to a different ontological category than the item belongs to, or (iii) would contradict a marked (— # or key X) cell. The extended Check 6 confirms that all 96 sentence interpretations — including those where interpretation equals the original sentence — are consistent with the formal context. Where

modifications are made, they uniformly apply correct EO term capitalisation, the established bracket convention (Constraint 8), or conceptual refinements that are directly supported by adjacent discourse sentences.

Absent-cell reasoning key patterns:

• Disjoint: Concrete thing / Conceptual thing (54-02 constraint) — excludes Conceptual characteristics from Concrete items and vice versa.

• Disjoint: Concrete thing / Abstract thing — excludes Abstract characteristics from all Concrete items.

• Disjoint & complete: in a mind / in reality — excludes 'in a mind' for all Concrete things; excludes 'in reality' for Conceptual and Abstract things.

• Disjoint & complete: Tangible concrete thing / Intangible concrete thing — excludes tangible characteristics from intangible items and vice versa.

• Disjoint & complete: Composite thing / Elemental thing — excludes Elemental sub-types from all Composite items.

• 'Can represent a Thing' is a characteristic of Conceptual things only; Concrete and Abstract things do not represent other Things.

• 'Cannot represent a Thing' (58-03) applies only to Abstract things.

• Abstract things are Elemental and thus excluded from all Composite sub-type characteristics.

Extraction: pdfplumber, 27 rows × 34 columns, all 913 non-placeholder cells enumerated prior to validation.